ICUBE DEVELOPMENT (CALGARY) LTD.

Timeclock Build and Install Guide

Step by step, from printing the case to a working clock on the wall. No previous experience needed. Follow every step in order.





A card clock you build, install and manage locally. No subscription, no per-employee fees and no cloud account.

Case v8.34 · Image bpi-zero-wbuild 3.13 · mk-timeclock Lite 1.9.5-RC3-cleanup27

Revised October 25, 2026

START HERE

Read this first

What you'll do

PART	WHAT HAPPENS	CHAPTER
A. Build	Print the case, solder the parts, prepare the memory card, put the parts in the case and connect the wires	1 to 5
B. Software	Start the clock, copy the timeclock software to it and install it	6 to 9
C. Finish	Create the first manager, test, close the case and hang it on the wall	10 and 11

How to read this guide

- **Bold words** are buttons, labels or messages you'll see.
- Text in a grey box like the one below is something you **type exactly**, including spaces and capital letters. Press **Enter** at the end of each line.

```
cd /root
```

- Do every step in order. Don't skip ahead.
- Tick the boxes ☐ as you go.

Safety

> **Never connect power until the guide tells you to**
>
> Wiring with power connected can destroy the board. Each time you change a wire, unplug the USB cable first.

- The card reader and display run on 3.3 V. Never connect either one to a 5 V pin.
- A soldering iron is about 350 °C. Use a stand, work on a heat-proof surface and keep the room ventilated.
- Touch a metal object before handling the boards, to discharge static. Hold boards by their edges.

You'll need some help with

- The **Wi-Fi name and password** the clock will use. It must be a **2.4 GHz** network.
- Access to your **router's admin page** (to find the clock's address), or someone from IT who has it.
- For the first clock, the **manager's name, work email** and a new card for them.
- Later, for email: the details of your company's **outgoing mail (SMTP) server**.

Attendance is stored on the clock itself. Extra (Secondary) clocks need a network connection so they can send punches to the Primary clock.

Shopping list

☐	PART	QTY	NOTES
☐	Banana Pi BPI-M2 Zero	1	Comes without pins (bare holes)
☐	microSD card	1	16 GB or larger, good brand
☐	2.42" SSD1309 OLED display, 128 × 64	1	The 7-pin module shown on the Components page
☐	RC522 (MFRC522) RFID reader kit	1	Includes a test card and key fob. Staff cards must be 13.56 MHz types that work with the RC522
☐	Active buzzer with mounting ears, 22.6 mm	1	2 wires (red and black). Must work from a 3.3 V signal
☐	USB-C breakout board (pads G, +D, −D, V)	1	The type shown on the Components page
☐	Wi-Fi antenna, 2.4 GHz, IPEX / u.FL plug	1	Flat stick-on PCB type
☐	Right-angle (90°) male pin header, 2.54 mm	1 strip	Enough for 40 (2 rows of 20) + 8 + 7 + 2 pins
☐	Female-to-female jumper wires (Dupont), 10 to 20 cm	14	Different colours make checking easier
☐	M2 × 5 mm self-tapping screws	8	OLED (4) and RFID reader (4)
☐	M2 × 6 mm self-tapping screws	6	Banana Pi (4) and USB-C board (2)
☐	M2 × 8 mm self-tapping screws	2	Buzzer
☐	M3 × 12 mm countersunk (flat head) self-tapping screws	4	Rear cover
☐	USB charger (regulated 5 V, 2 A or more) and USB-A to USB-C cable	1	A good-quality phone charger with the big USB-A socket is ideal
☐	2 wall screws and anchors	2	Pan head. Head no wider than 8 mm, shank no thicker than 4.3 mm
☐	Heat-shrink tube, small	a few cm	For the buzzer wire joins

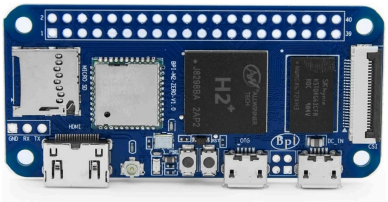
Tools

3D printer and ASA or PLA filament · soldering iron, solder and flux · side cutters · small Phillips screwdriver · tweezers · a computer with a microSD card reader · a spirit level, drill and pencil for wall mounting.

Screw lengths come from the case design. If a screw feels like it's bottoming out, stop. Don't force it.

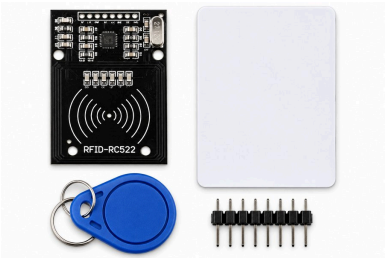
Components

Check you have each of these and that they look like the photos.



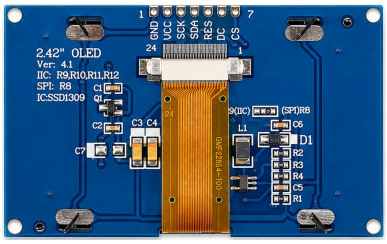
Banana Pi BPI-M2 Zero

The small computer that runs the clock. The 40 holes along one edge are the "header". You'll solder pins into them.



RC522 card reader

Reads the staff cards. Its 8 holes are labelled SDA, SCK, MOSI, MISO, IRQ, GND, RST, 3.3V. Keep the test card and fob. Use right-angle pins, not the straight ones in the kit.



OLED display, back

7 holes labelled 1 GND, 2 VCC, 3 SCK, 4 SDA, 5 RES, 6 DC, 7 CS. It needs a small change before use (Chapter 2). Board versions vary, so go by the printed labels.



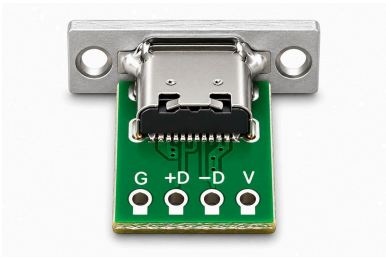
OLED display, front

The glass shows through the case window. Don't press on the glass or bend the orange ribbon.



Buzzer

Beeps on every tap. Red wire is + (plus), black is − (minus). The two ears take screws.



USB-C board

Where the power cable plugs in. Only the **V** and **G** holes are used. +D and −D stay empty. Go by the printed letters, not their position in the photo.

CHAPTER 1

Print the case

Download the two case files from the project's GitHub page (links on the last page of this guide):

☐ FILE	WHAT IT IS
☐ `front.stl`	The front of the case
☐ `back.stl`	The rear cover

The file `assembled.stl` shows how the two parts fit together. You don't print it.

Material

ASA is recommended. PLA also works. No supports are needed.

Slicer settings

1. Open your printer's slicer (for example Bambu Studio, PrusaSlicer or Cura) and load the file.
2. Place the **front** with its **outside face on the bed**. Place the **rear cover** with its **flat outside face on the bed**.
3. Choose your printer's own profile for the material you're using. Then set: nozzle **0.4 mm**, layer height **0.24 mm**, **3 walls**, **20 % infill**, **supports off**. These are a starting point.
4. Slice and print.

After printing

- ☐ Pick off any strings of plastic inside the **display window**, the **buzzer holes** and the **screw holes**.
- ☐ Check the cover fits snugly over the front's rim all the way round.

> **Use the GitHub STL files**
>
> The standard build uses the STL files exactly as published. A SCAD source file is also on GitHub for anyone who needs to regenerate a part in OpenSCAD (render with F6, then export as STL). You don't need it for a normal build.

CHAPTER 2

Solder the parts

> **Right-angle pins only**
> The pins must be right-angle (bent 90°) pins, 2.54 mm. Straight pins stick up too far and the case won't close.

1. Banana Pi: 40 pins

1. Cut a 2 × 20 piece of right-angle header (or two 1 × 20 strips).
2. Push the short ends into the 40 holes from the **top** of the board (the side with the big black chips). The long ends must point **away from the board**, out past its edge.
3. Turn the board over and solder every pin on the underside. Each joint should be a small shiny cone. Make sure no blob of solder touches a neighbouring pin.

2. RC522 reader: 8 pins

Push the short ends into the 8 holes from the **plain back** of the reader (the side without the printed antenna coil), long ends pointing **away from the board**. Solder on the antenna side.

3. OLED display: 7 pins

Push the short ends in from the **back** (the side with the printing), long ends pointing **inward, over the back of the board**, like the reference photo. There's no room above the top edge of the display inside the case. Keep the iron away from the glass and the orange ribbon.

4. USB-C board: 2 pins

Solder right-angle pins into **V** and **G** only, so they point along the board, away from the socket. Go by the letters printed next to the holes.

5. Buzzer wires

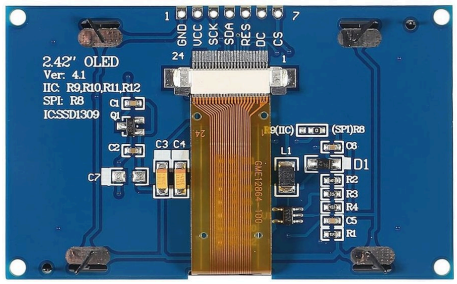
Cut one end off two jumper wires. Strip about 5 mm from the buzzer wires and the jumper wires, twist red to one and black to the other, solder, and cover each join with heat-shrink. No bare metal should show. Label or remember which is red (+).

> **Before you go on**
> Look at every solder joint. No solder should connect two neighbouring pins.

Change the display to I2C mode

The display comes set for a different connection type (SPI). The clock needs it set to **I2C** (printed **IIC** on the board). You change it with five tiny positions on the back, each marked with an R number.

> **R9 to R12 are under the orange ribbon**
>
> You have to take the ribbon out of its connector to reach them, then put it back afterwards. Take a photo of the ribbon and connector first, so you know how it goes back.



The ribbon plugs into the connector at the top, under pins 1 to 7. The printing "IIC: R9, R10, R11, R12 / SPI: R8" is at the top left.

1. Take the ribbon out

1. Lay the display glass-side down on a soft cloth.
2. At the connector, gently flip up the small **locking bar** with a fingernail or plastic tool. Never pull the ribbon while the bar is closed.
3. Slide the ribbon straight out. If it's taped to the board, peel the tape off slowly. Lay the ribbon aside **without folding or creasing it**. It stays attached to the glass.

2. Change the five positions

☐	POSITION	WHAT TO DO
☐	R8	Remove the tiny 0 Ω resistor (a small part marked "0"), leaving the pads open. Add a little fresh solder, heat both ends at once and slide it off with tweezers.
☐	R9	Solder that removed resistor here, or join the two pads with a blob of solder
☐	R10, R11, R12	Join each pair of pads with solder: a little flux, then touch the iron with a small amount of solder across both pads. You want one shiny bridge per pair.

Check with a magnifier that no solder reaches a neighbouring pad. Board versions vary, so go by the R labels printed on your board. No other change is needed. The clock looks for the display at address 0x3C.

3. Put the ribbon back

1. Slide the ribbon back in **the same way round** as in your photo, all the way in and square to the connector.
2. Press the locking bar down until it's flat. Tug the ribbon very lightly: it shouldn't move.

> **Small parts**
>
> The parts and the ribbon are delicate. Use good light and tweezers. If you're not comfortable, ask someone experienced to do this step.

CHAPTER 3

Prepare the memory card

Do this on your computer, **before** you put the parts in the case. The card has to go into the Banana Pi before the Banana Pi is screwed in, because the buzzer blocks the card slot once everything is fitted.

1. Download the clock's system

1. Go to **github.com/mkinderwater/bpi-zero-wbuild/releases/tag/3.13**.
2. Scroll down to **Assets**. Click **bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz** (about 148 MB) to download it.
3. Don't open or unzip it.

2. Write it to the card

1. Install **balenaEtcher** from **balena.io/etcher** and open it.
2. Put the microSD card in your computer's card reader.
3. Click **Flash from file** and choose the **.img.gz** file you downloaded.
4. Click **Select target** and tick your microSD card. Check the size matches your card, so you don't erase the wrong drive. Everything on the card will be erased.
5. Click **Flash!** and wait until it says **Flash Complete**.

> **Windows: if a message says "You need to format the disk", click Cancel**
> Formatting erases the card and you'd have to start again. Windows may show this more than once. Always click **Cancel**.

3. Open the settings file

1. Take the card out and put it back in.
2. A drive called **BPIWBUILD** appears. Open it.
3. Open the file **CONFIG.TXT**:
   Windows: right-click it and choose **Open with → Notepad**.
   Mac: open it with **TextEdit**. If you see formatting buttons, choose **Format → Make Plain Text**.

Fill in the settings

Scroll to the bottom of CONFIG.TXT. Lines starting with **#** are only notes, so leave them alone. Change only the **six lines** at the bottom so they look like this, with your own details:

```
SSID=OfficeWiFi
PSK=wifi-password-here
COUNTRY=CA
HIDDEN=false
TIMEZONE=America/Edmonton
ROOT_PASSWORD=Choose-A-Strong-Password-26
```

LINE	WHAT TO PUT AFTER THE =
SSID	The Wi-Fi network name, exactly as it appears on a phone (capitals matter)
PSK	The Wi-Fi password (8 to 63 characters)
COUNTRY	Two-letter country code: **CA** for Canada, **US** for the United States
HIDDEN	**false**, unless IT tells you the network is hidden (then **true**)
TIMEZONE	Your area, copied from the list in the file, for example America/Edmonton, America/Vancouver or America/Toronto
ROOT_PASSWORD	A new password for the clock, 8 to 64 characters. **Required.** Without it you can't get into the clock

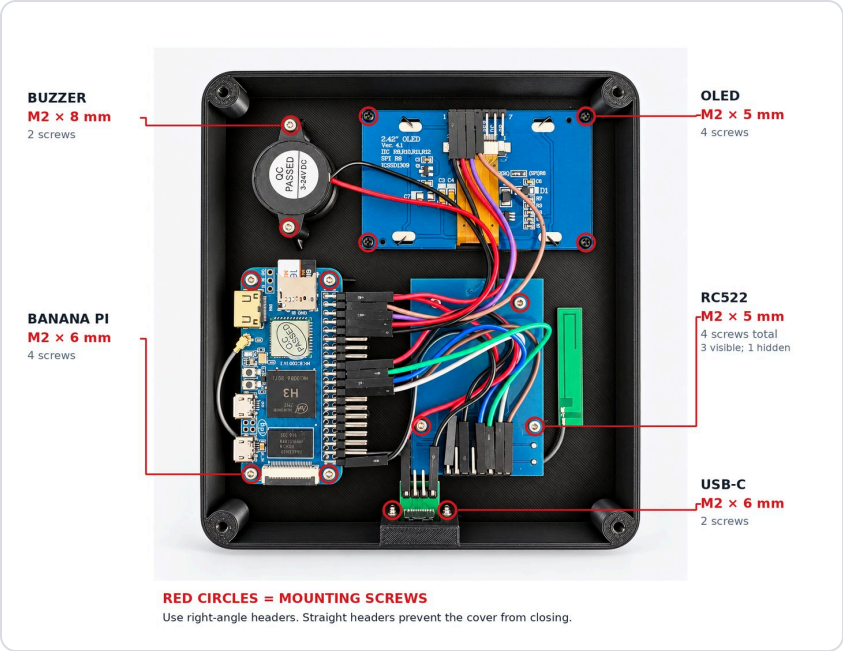
- **No spaces** before or after the **=**, and **no quote marks**.
- **Write down the ROOT_PASSWORD now** and keep it safe. The clock deletes it from this file after it starts, and you'll need it in Chapter 7.

1. Save the file (**Ctrl+S** on Windows, **Cmd+S** on Mac).
2. Eject the card properly: Windows, right-click the drive and choose **Eject**. Mac, drag it to the Trash.
3. Keep the card somewhere safe. It goes into the Banana Pi in Chapter 4.

CHAPTER 4

Put the parts in the case

Lay the front face-down on a soft cloth, open back facing you. Match each part to the photo. Turn the screws in gently and stop when they're snug, because they cut their own thread in the plastic.



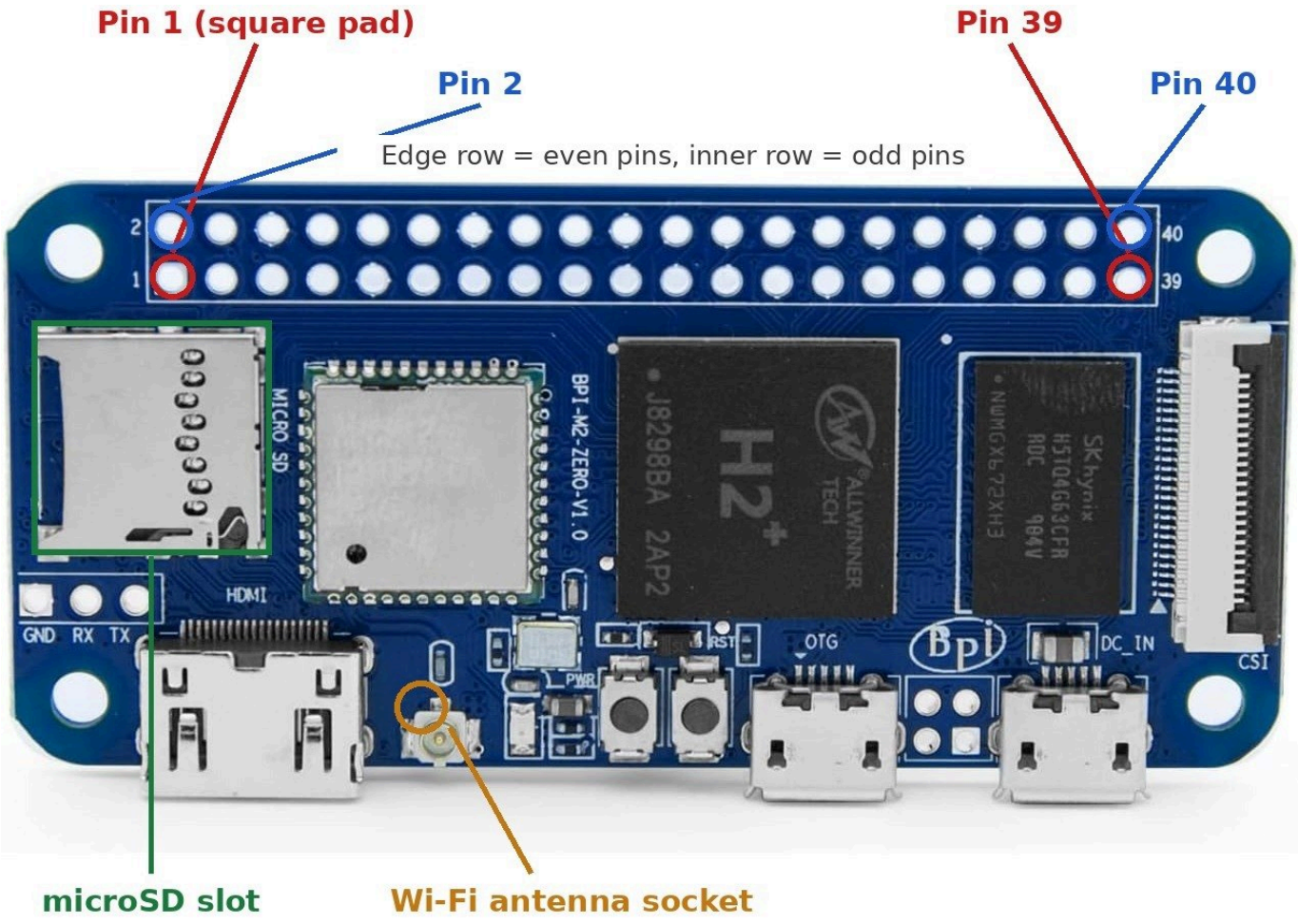
From the open back. Red circles are the mounting screws. One card reader screw is hidden by the wiring.

1. ☐ **USB-C board** (bottom edge, middle). Seat it in the bottom slot so the socket faces down. Fix it with **2 × M2 × 6 mm**.
2. ☐ **Display** (top right). Press the glass into the window from inside. Fix it with **4 × M2 × 5 mm** into the 4 corner holes.
3. ☐ **Card reader** (below the display). Antenna side facing the front of the case. Fix it with **4 × M2 × 5 mm**.
4. ☐ **Buzzer** (top left). Drop it into the round ring. Fix it with **2 × M2 × 8 mm** through its ears.
5. ☐ **Banana Pi** (bottom left). **First** slide the memory card from Chapter 3 into its slot, gold contacts facing the board, until it clicks. Then fit the board chips facing you, microSD slot at the top, and fix it with **4 × M2 × 6 mm**.
6. ☐ **Wi-Fi antenna.** Press its tiny plug onto the gold antenna socket on the Banana Pi, near the HDMI port, until it clicks (Chapter 5 shows where). Stick the antenna **upright, to the right of the card reader**, and run its lead around the bottom edge.

> **Memory card in before the Banana Pi**
> The buzzer blocks the card slot once the Banana Pi is in. Leave the rear cover off for testing.

CHAPTER 5

Find the pins on the Banana Pi



The board has 1, 2, 39 and 40 printed next to the header

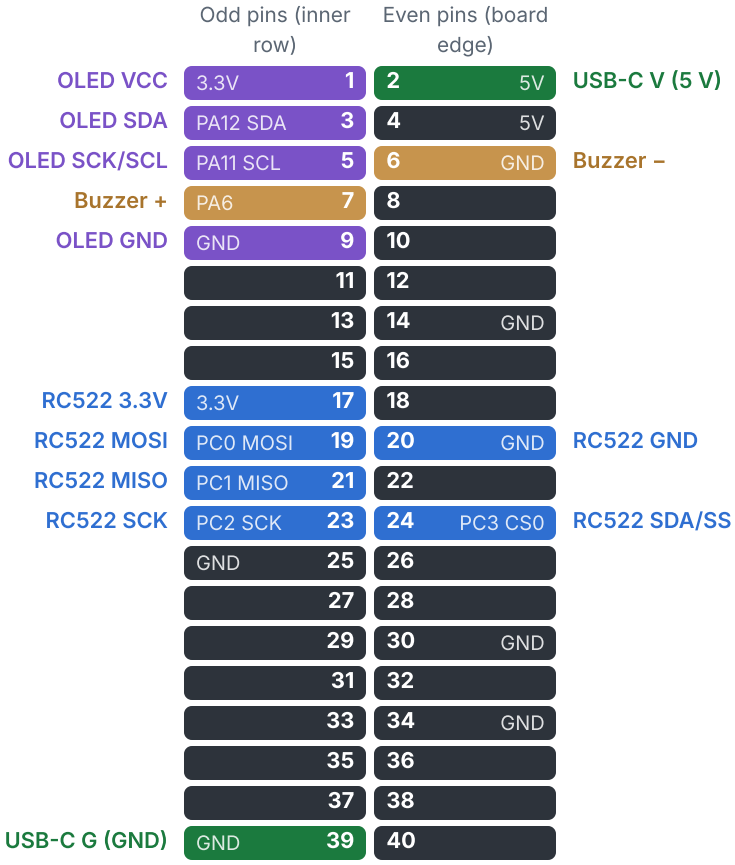
- Pin **1** has a **square** pad and is at the end nearest the microSD slot.
- The row along the **edge of the board** has the **even** pins: 2, 4, 6 … 40.
- The **inner** row has the **odd** pins: 1, 3, 5 … 39.
- Always read the pins with the **chip side** of the board facing you, not from underneath or from the ends of the bent pins.
- To find a pin, count along its row in twos from the nearest printed number. For example, pin 24 is the 12th pin along the edge row from pin 2.

> **Tip**
> Before wiring, put a small dot of marker next to pins 1, 9, 17, 24 and 39 as landmarks.

Connect the wires

The USB cable must be **unplugged**. Push each jumper firmly onto its pin. Do them in this order and tick each one.

☐ #	FROM (LABEL ON THE PART)	TO BANANA PI PIN
Card reader (RC522)		
☐ 1	SDA	**24**
☐ 2	SCK	**23**
☐ 3	MOSI	**19**
☐ 4	MISO	**21**
☐ 5	GND	**20**
☐ 6	3.3V	**17** (never 2 or 4)
	IRQ and RST	Leave empty
Display (OLED)		
☐ 7	1 GND	**9**
☐ 8	2 VCC	**1**
☐ 9	3 SCK	**5**
☐ 10	4 SDA	**3**
	5 RES, 6 DC, 7 CS	Leave empty
Buzzer		
☐ 11	Red (+)	**7**
☐ 12	Black (–)	**6**
USB-C board		
☐ 13	V	**2**
☐ 14	G	**39**



Every other pin stays empty.

Same letters, different jobs

The card reader's **SDA** (pin 24) and the display's **SDA** (pin 3) are different connections. Wire each one exactly as the table says.

Check the wiring

Mistakes here can destroy parts, so check slowly. Ask a second person to check too.

☐	CHECK
☐	The memory card is in the Banana Pi's slot.
☐	You count exactly **14 wires** on the Banana Pi pins, plus the thin antenna lead.
☐	**Pin 2** goes only to the USB-C board's **V**. Nothing else is on pins 2 or 4 (they carry 5 V).
☐	The card reader's **3.3V** wire is on **pin 17**, and the display's **VCC** wire is on **pin 1**.
☐	All three ground wires are right: card reader GND on **20**, display GND on **9**, USB-C G on **39**.
☐	The buzzer's **red** wire is on **pin 7** and **black** on **pin 6**.
☐	Each plug sits fully on one pin, not between two pins or off the end of a row.
☐	No loose wire strands, and no bare metal or solder touches anything it shouldn't.
☐	The wires lie flat and don't press on the display ribbon.

> **Still don't plug in power**
> Power goes on in Chapter 6.

CHAPTER 6

Start the clock

1. ☐ Plug the USB-A to USB-C cable into the charger, then into the clock's USB-C socket on the bottom of the case.
2. ☐ A small **red light** on the Banana Pi comes on. If nothing lights up, unplug and recheck wires 13 and 14.
3. ☐ **Wait 10 minutes.** On its first start the clock sets itself up and joins the Wi-Fi. The display stays blank until the timeclock software is installed in Chapter 9. That's normal.

Find the clock's address

Your computer needs the clock's network address (its **IP address**, four numbers like 192.168.1.50).

1. Open your router's admin page (or ask IT) and look at the list of connected devices.
2. Find the new device that appeared in the last few minutes. Its name is a mix of letters and numbers.
3. Write down its IP address: _____ . _____ . _____ . _____
4. In the router, choose **Reserve** (also called static or fixed IP) for this device, so its address never changes.

A phone app such as Fing can also list devices on your Wi-Fi. If you connect a monitor to the Banana Pi's mini-HDMI port, the screen shows **Wi-Fi: wlan0 MAC: … IPv4: …** with the address.

> **Not on the list after 15 minutes?**
>
> Unplug the power. Unscrew the Banana Pi (4 screws) and lift it out gently, keeping the wires on. Take the card out, put it in your computer and check CONFIG.TXT: the Wi-Fi name and password exactly right, a 2.4 GHz network, and ROOT_PASSWORD filled in. Fix it, save, eject, put the card back in the Banana Pi, screw the board back in and try again.

CHAPTER 7

Connect to the clock

You'll type commands to the clock from your computer. This is called **SSH**. It's built into Windows 10, Windows 11 and Mac.

1. Open a command window:
   Windows: click **Start**, type **PowerShell** and press Enter.
   Mac: open **Applications → Utilities → Terminal**.
2. Type this, using your clock's IP address instead of 192.168.1.50, and press Enter:

   ```
   ssh root@192.168.1.50
   ```
3. The first time, it asks **Are you sure you want to continue connecting (yes/no)?** Type **yes** and press Enter.
4. It asks for a **password**. Type the ROOT_PASSWORD from Chapter 3 and press Enter. **Nothing appears on screen while you type the password.** That's normal. Just type it and press Enter.
5. You're in when the line ends with **#**, for example **root@...:~#**.

Check it's the right system. Type:

```
cat /etc/bpi-zero-wbuild-release
```

The answer should include **3.13-trixie-minimal-hw**.

> **Copy and paste**
> You can copy commands from this guide. To paste into PowerShell, right-click. On Mac, press Cmd+V. Check the pasted text before pressing Enter.

To leave the clock's command window later, type **exit**. Keep it open for now.

CHAPTER 8

Copy the software to the clock

Download it first

1. Go to **github.com/mkinderwater/mk-timeclock/releases**.
2. Under the latest release, open **Assets** and download the ZIP file, for example **mk-timeclock-lite-RC3-cleanup27.zip**. Don't unzip it.

This guide uses cleanup27 in every example. If your download has a different name, use that name instead wherever you see cleanup27.

Now pick **one** of these ways to copy it to the clock.

Windows: WinSCP (easiest)

1. Install **WinSCP** from **winscp.net** and open it.
2. In the login box: File protocol **SFTP**, Host name = the clock's IP address, Port **22**, User name **root**, Password = your ROOT_PASSWORD.
3. Click **Login**. If asked about a host key, click **Accept**.
4. The right side shows the clock. Make sure it's in the folder **/root**.
5. Drag the zip file from the left side (your computer) to the right side.

Mac, or Windows PowerShell

1. Open a **new** Terminal or PowerShell window (keep the SSH one open).
2. Go to the folder with the zip file. If it's in Downloads, type:

```
cd Downloads
```

3. Then type (using your clock's IP address):

```
scp mk-timeclock-lite-RC3-
cleanup27.zip
root@192.168.1.50:/root/
```

4. Type the ROOT_PASSWORD (invisible again) and press Enter. Wait for **100%**.

CHAPTER 9

Install the timeclock software

Go back to the SSH window from Chapter 7 (the one ending in **#**). Type each line, press Enter, and wait for the **#** to come back before typing the next.

☐ TYPE THIS	WHAT IT DOES
☐ `cd /root`	Goes to the folder you copied the zip into
☐ `unzip mk-timeclock-lite-RC3-cleanup27.zip`	Unpacks the software
☐ `cd mk-timeclock-lite-RC3-cleanup27`	Goes into the unpacked folder
☐ `./install-master.sh`	Installs the clock

While it installs

The installer downloads what it needs, builds the software and sets up the clock's services by itself. Lots of text scrolls past and it can take several minutes, longer on slow internet.

Don't close the window or unplug the clock. It's finished when the **#** comes back and the last lines don't show an error.

If it ends with an error

Read the last lines of text: they say what went wrong. Check **If something goes wrong** at the back of this guide before running the installer again.

More than one clock?

The first clock is the **Primary**. It keeps the attendance records. Extra clocks are **Secondary** clocks and send their punches to the Primary over the network. To add one, follow the **Manager Guide**.

CHAPTER 10

Create the first manager

Use a browser on a computer connected to the same network as the clock.

1. Go to **https://192.168.1.50/setup** (your clock's IP address).
2. The browser shows a **security warning**. That's expected at this stage, because the clock uses a temporary certificate.
   Chrome or Edge: click **Advanced**, then **Proceed to ...** or **Continue to ...**
   Safari: click **Show Details**, then **visit this website**.
   Firefox: click **Advanced**, then **Accept the Risk and Continue**.
3. Fill in the manager's name, work email, position and a password of **12 or more characters**. Submit the form.
4. The page says **Tap the new manager card on the clock within 30 seconds**. Tap a **new, unused card** on the clock's tap area. That card is now the manager card.
5. Sign in at **https://192.168.1.50/login**.

> **Manager card and attendance card are different**
> Keep the manager card only for manager sign-in. Give the manager a second card for their own attendance.

What's next

- The **Manager Guide** covers employees, schedules, email, payroll and adding more clocks.
- Give staff the **Employee Guide**.

For IT, later

So email links work without warnings, install a trusted certificate under **Settings → HTTPS certificate**. Set up **Settings → Outgoing email** with your SMTP server so the clock can send email.

CHAPTER 11

Test, close and hang it

Test



☐ **Display works**
Shows the date, time and TAP YOUR CARD.



☐ **Manager card**
Tap it: HELLO MANAGER, and a beep.



☐ **Test card from the kit**
Tap it: INVALID CARD, NOT RECORDED. That's correct, because it isn't registered.

If the clock says **CLOCK NOT SYNCED**, it can't get the correct time from the internet yet. Wait 5 minutes and try again, then ask IT.

Close the case

Only after all three tests pass.

1. Unplug the power.
2. Check no wire is pinched at the edge. Fit the rear cover over the rim, with the vent slots behind the Banana Pi and the two keyholes at the top.
3. Fix it with **4 × M3 × 12 mm** flat-head screws. They sit flush with the back.

Hang it on the wall

1. Print the separate **wall mounting template** (link on the last page) at **100 % / Actual size**, not "fit to page". Check the 80 mm line measures 80 mm with a ruler.
2. Tape it to the wall at a comfortable height, level, near a power socket. Keep away from large metal surfaces and direct sunlight.
3. Drill at the two crosses, fit the anchors and screw in the two screws, leaving the heads about **3 mm** out from the wall.
4. Hold the clock so the screw heads go through the round part of the keyholes, then slide it **down** until it stops.
5. Plug in the USB cable from below and wait for **TAP YOUR CARD**.

HELP

If something goes wrong

WHAT YOU SEE	WHAT TO DO
No red light on the Banana Pi	Unplug. Check wires 13 (V to pin 2) and 14 (G to pin 39). Try another charger and cable.
The clock never appears on the router list	Check CONFIG.TXT (Chapter 3; see Chapter 6 for how to get the card out): Wi-Fi name and password exactly right, 2.4 GHz network, ROOT_PASSWORD filled in. Wait 10 minutes after power on. Check the antenna plug clicked onto its socket.
Permission denied when typing the password	The user name must be root. Retype the ROOT_PASSWORD carefully (it's invisible as you type). Capitals matter.
ssh is not recognized (Windows)	Use WinSCP for copying, and its **Commands → Open in PuTTY** option, or ask IT to enable OpenSSH Client.
No such file or directory after unzip or cd	The zip isn't in /root, or its name is different. Repeat Chapter 8, then type `ls /root` to see what's there and use that exact name.
The installer says **BPI image marker missing** or **Missing /dev/...**	The card wasn't written with the 3.13 system. Repeat Chapter 3 with a fresh download (unscrew the Banana Pi to get the card out).
Permission denied running ./install-master.sh	Type `cd /root`, delete the folder with `rm -r mk-timeclock-lite-RC3-cleanup27`, and unzip again with the unzip command.
Display stays blank after install	Wires 7 to 10 (pins 9, 1, 5, 3) and the I2C change in Chapter 2. Check the display ribbon is fully back in its connector, square, with the locking bar closed, and that no solder touches a neighbouring pad.
Cards are never read (no beep, nothing on screen)	Wires 1 to 6. 3.3V must be on pin 17. Antenna side facing the front. Use 13.56 MHz cards that work with the RC522.
No beep	Buzzer red on pin 7, black on pin 6.
Browser warning on the clock's page	Normal until IT installs a certificate (Chapter 10).

> **Write this down for each clock**
>
> Primary or Secondary · location · IP address · date installed · software version. Keep the ROOT_PASSWORD in a safe place.

REFERENCE

Where to find everything

Everything for the clock is on GitHub. Download the newest version of each file when you start a build.

WHAT	WHERE
Timeclock software (ZIP)	github.com/mkinderwater/mk-timeclock/releases
Clock system image 3.13	github.com/mkinderwater/bpi-zero-wbuild/releases/tag/3.13
Case front	github.com/mkinderwater/mk-timeclock/blob/main/front.stl
Case rear cover	github.com/mkinderwater/mk-timeclock/blob/main/back.stl
Assembled model (view only)	github.com/mkinderwater/mk-timeclock/blob/main/assembled.stl
Case SCAD source	github.com/mkinderwater/mk-timeclock/blob/main/bpi_rfid_timeclock_v8_34_two_piece.scad
Wall mounting template	github.com/mkinderwater/mk-timeclock/blob/main/Timeclock-Wall-Mounting-Template.pdf
Manager Guide	github.com/mkinderwater/mk-timeclock/blob/main/Timeclock-Manager-Guide.pdf
Employee Guide	github.com/mkinderwater/mk-timeclock/blob/main/Timeclock-Employee-Guide.pdf
Source code and license	github.com/mkinderwater/mk-timeclock

Version covered by this guide

Hardware	Banana Pi BPI-M2 Zero
Case	v8.34
System image	bpi-zero-wbuild 3.13 (Debian Trixie, minimal-hw)
Software	mk-timeclock Lite 1.9.5-RC3-cleanup27
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